

Quantitative Analysis of Media Bias and Stock Price Dynamics: The 2020 Shock

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Abstract—

I. INTRODUCTION

II. RELATED WORK

III. DATA COLLECTION

A. Corpus and Coverage

Our object of study is the tone of the news that surrounds large United States firms, so the corpus has to begin as close as possible to the full stream of that coverage. We assemble it from MediaCloud, a service that continuously indexes the output of national news outlets, and we query its United States national collection one firm at a time over the period 2015 to 2025. Each query returns the news items that name the firm, each carrying a publication date, a source outlet, and a headline. We keep the headline as the unit of analysis, because the headline is the one part of a news item written to compress the story's angle into a single line, and it is therefore where a firm's coverage wears its stance most plainly. Cast this way the net is deliberately wide: seeded with roughly 250 large-capitalization United States firms, the raw pull returns 6,284,404 headlines, a stream large enough that no single outlet or stretch of time dominates it.

A corpus of this size invites an obvious temptation, which is to make it larger still, and we tested whether a second aggregator, GDELT, could extend the coverage we had already gathered. It could not. The headlines GDELT returned were of visibly poorer quality, often truncated or malformed and heavily duplicated across near-identical records, so folding them into the corpus would have thinned a clean signal with noise rather than deepening it. We keep the study on the MediaCloud stream, whose headlines are consistently well formed and attributed, and treat that stream as the raw material for everything that follows.

One feature of that raw material shapes every later choice. Coverage is spread across firms with extreme inequality: a small number of the most heavily reported names account for the bulk of all headlines, while a long tail of firms surfaces in the news only occasionally. This unevenness is the first

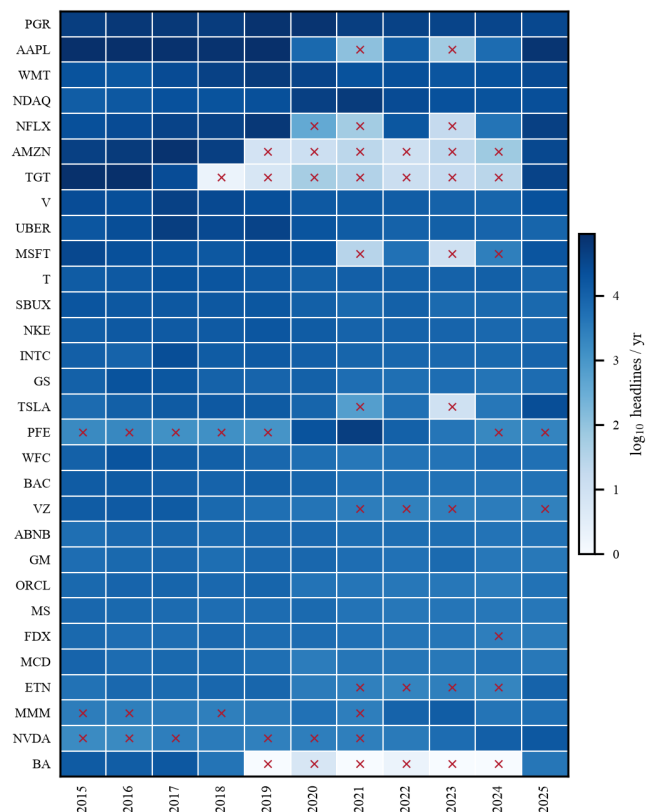


Fig. 1. Yearly headline coverage for each candidate firm, shaded by $\log_{10}$ headlines per year. A cross marks any year in which a firm sits at or below the three-thousand-headline floor; a firm crossed in more than three of the eleven years is set aside as too intermittently covered to support a continuous series.

thing the data asks us to confront once we begin turning a raw stream of headlines into a panel we can measure.

B. Data Preprocessing

The skew is not a nuisance to be smoothed away; it decides which firms can be studied at all. A daily, firm-level measure

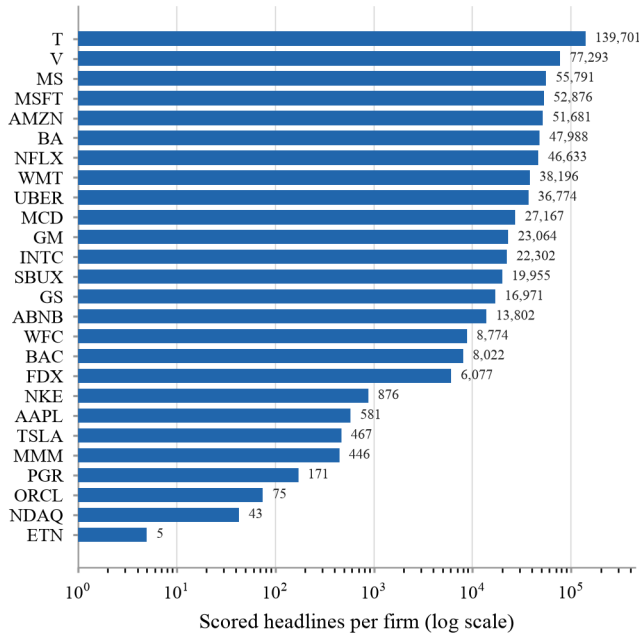


Fig. 2. Scored headlines per firm in the final corpus (26 firms, log scale). Coverage spans more than four orders of magnitude, from AT&T at 139,701 down to Eaton at 5, which the firm fixed effects in the panel regressions absorb.

of tone is only as steady as the coverage beneath it, and a firm the news names a handful of times a month cannot carry a daily series that means anything. We therefore keep the firms the press reports on heavily and let the thin ones go. Ranking the universe by the volume of coverage each firm attracts and holding onto the most reported names concentrates the corpus onto its densely covered head, some 4,706,589 headlines, and leaves behind a long tail of firms that never generated enough news to measure.

Volume in a single year is not enough on its own, because coverage also has to persist across the whole decade: a firm that falls quiet for years leaves stretches of time that no series can span. We therefore hold each remaining firm to a floor of more than three thousand headlines in a year, and set aside any firm that slips to or below that floor in more than three of the eleven years from 2015 to 2025, since a firm that runs thin that often cannot anchor a continuous weekly or daily series. Figure 1 maps yearly coverage firm by firm and marks the years that fall below the floor; the firms carried forward are those whose rows stay dark almost throughout, while the ones that flicker pale in year after year are the ones the rule removes.

Two kinds of contamination still sit inside what survives. A great many headlines name a firm only in passing, in a list or an aside, while the story itself is about something else; because a headline written about the firm almost always carries the firm’s name or its ticker symbol, we keep only the headlines that do and discard the rest. The same wire report, in turn, is republished word for word across dozens of outlets, so

we collapse these near-identical copies to a single record per firm. Table I traces the corpus through these passes, from the full scrape down to what remains: 695,731 scored headlines across 26 firms, each attached to a firm the news followed closely and each naming that firm in its title. Even within this retained set coverage stays strikingly unequal, spanning more than four orders of magnitude from the most reported firm to the least (Figure 2); the estimation later leans on firm-level effects precisely so that these permanent differences in coverage volume cannot masquerade as differences in tone.

TABLE I
FROM THE FULL NEWS SCRAPE TO THE CORPUS THE STUDY RUNS ON.

Stage	Headlines	Firms
Full MediaCloud scrape	6,284,404	250
Most-covered firms retained	4,706,589	30
Name or ticker in title, de-duplicated	695,731	26

IV. METHODOLOGY

A. Relevance Classification

B. A Panel Design for the 2020 Break

With a daily, firm-level stance series in hand and a matching series of daily stock returns, the first question the 2020 shock raises is the plainest one: did either series step to a new level once the shock arrived? The tempting way to answer it is to average each series over the years before 2020 and over the years after, and read off the difference. That answer would mislead, and seeing why fixes the design for both questions that follow.

Two forces move a firm’s coverage and its returns that have nothing to do with the shock. Firms differ from one another in ways that hold across the whole sample: some are simply covered more, and more warmly, than others, and some earn higher average returns than others. And on any single day every firm is pushed in the same direction by the news cycle and by the market as a whole. A raw before-and-after average folds both of these into what it labels the “2020 effect”: a shift in which firms happen to be in the news, or a run of shared bad days, would surface as a break that owes nothing to the shock itself.

A panel regression with two sets of fixed effects strips out exactly these two confounders. Firm effects pin each firm to its own baseline, so the comparison is between a firm and its own past rather than between one firm and another; time effects hold constant whatever is common to every firm on a given day, or in a given month, so a market-wide swing cannot be read as a firm-level break. What remains for a post-2020 indicator to explain is the within-firm change net of common shocks, which is exactly the quantity the shock question asks about. This is why we estimate the break inside a two-way fixed-effects regression rather than from raw means, from a single firm’s time series, or from an event study around one date: only the panel holds the standing differences between firms and the shared movements across them fixed at the

same time. The estimate is credible under three assumptions the setting makes reasonable: that, absent the shock, the pre- and post-periods would have shared a common trend; that the timing of the break is set by the pandemic's onset rather than by anything in a firm's own coverage or returns; and that, after the within-transformation, what is left is uncorrelated with the post indicator. To guard the last of these we report HC1-robust standard errors throughout, and, where several firms are covered on the same day, standard errors clustered by day.

C. RQ1: Did Media Stance Shift After the Shock?

The outcome for this question is the tone of a firm's coverage on a day. Each retained headline is read by a target-dependent sentiment model that reports, for the firm named in it, how likely its tone toward that firm is positive, negative, or neutral; write these three probabilities as p_a^+ , p_a^- , and p_a^0 for headline a , with $p_a^+ + p_a^- + p_a^0 = 1$. We collapse them into a single signed score, $p_a^+ - p_a^-$, which runs from -1 for wholly unfavorable coverage to $+1$ for wholly favorable coverage and sits near zero when the headline is even-handed or the model is unsure. A firm's stance on a day is the average of these scores over the headlines it drew that day,

$$\text{bias}_{it} = \frac{1}{N_{it}} \sum_{a \in A_{it}} (p_a^+ - p_a^-), \quad (1)$$

where A_{it} is the set of retained headlines about firm i on day t , $N_{it} = |A_{it}|$ is that day's coverage volume, and $\text{bias}_{it} \in [-1, 1]$ is the average tone toward the firm.

We ask whether this daily stance moved to a new level after the shock. The regression places stance on the post-2020 indicator alongside firm and daily time effects and the day's coverage volume,

$$\text{bias}_{it} = \alpha_i + \delta_t^{\text{day}} + \beta \text{Post}_t + \gamma \text{vol}_{it} + \varepsilon_{it}. \quad (2)$$

Here $\text{Post}_t = 1[t \geq 2020-01-01]$ switches on for every day from January 2020 onward, and its coefficient β is the quantity of interest; α_i are the firm effects and δ_t^{day} the daily effects. The control $\text{vol}_{it} = N_{it}$ enters because a daily average tightens mechanically as more headlines are folded into it, so holding the count fixed keeps a genuine change in tone from being mistaken for a change in the sheer amount of coverage. We test

$$H_0^{(1)} : \beta = 0 \quad \text{against} \quad H_1^{(1)} : \beta \neq 0,$$

estimating (2) by the within-transformation with HC1-robust standard errors, and, since several firms can be covered on the same day, reporting day-clustered standard errors alongside them.

D. RQ2: Did Firm-Level Returns Shift After the Shock?

The second question asks the same of prices. The outcome is firm i 's daily log return, $\text{return}_{it} = \ln P_{it} - \ln P_{i,t-1}$, where P_{it} is the firm's closing price on day t ; we draw the daily closing prices, the S&P 500 index, and the VIX volatility index from `yfinance`. Log returns are additive over time and roughly symmetric, which suits a linear model. Returns carry a large

common component—when the market rises, most firms rise with it—so the design must strip that component out before it can see a firm-level break. Monthly time effects absorb the broad market regime, and within each month the S&P 500 return and the VIX account for the day-to-day market swings and the level of volatility. The regression is

$$\text{return}_{it} = \alpha_i + \delta_t^{\text{month}} + \beta \text{Post}_t + \gamma_1 \text{sp500_ret}_t + \gamma_2 \text{vix}_t + \varepsilon_{it}, \quad (3)$$

with Post_t as before, α_i the firm effects, δ_t^{month} the monthly effects, and $(\text{sp500_ret}_t, \text{vix}_t)$ the market controls. We again test the shock,

$$H_0^{(2)} : \beta = 0 \quad \text{against} \quad H_1^{(2)} : \beta \neq 0,$$

estimating (3) by the within-transformation with HC1-robust standard errors. Table II gathers every quantity that enters the two regressions, so that the returns model in particular is read as the post indicator and the market controls acting together rather than any one of them alone.

E. RQ3:

V. RESULTS

A. RQ1: No Detectable Shift in Media Stance

Estimating (2) on the 27,627 firm-day stance observations—26 firms across 3,982 trading days—returns the coefficients in Table III(a). The post-2020 coefficient is $\hat{\beta} = +0.011$ with an HC1 standard error of 0.030 ($p = 0.71$), and clustering by day barely disturbs it ($\text{SE} = 0.030$, $p = 0.72$), so the within-day dependence between firms is not hiding a result. We cannot reject $H_0^{(1)}$: once each firm is set against its own history and the day's common movement is held fixed, the tone of coverage does not step to a new level after the shock. The volume control behaves exactly as its purpose demands—its coefficient is -0.0038 ($p < 0.001$), so the days on which a firm is most heavily covered are the days its average tone is pulled back toward neutral, the mechanical averaging effect the control exists to absorb. The within R^2 of 0.002 is what one expects when a single firm's day-to-day tone is dominated by the noise of individual headlines. A null here is not a failure to find something; it says the shock left no common, firm-level mark on tone once the news cycle's own daily swings are taken out.

B. RQ2: No Detectable Shift in Firm-Level Returns

The same estimation on returns, (3) over 20,537 firm-day observations, tells the matching story (Table III(b)). The post-2020 coefficient is all but zero, $\hat{\beta} = -0.0001$ ($\text{SE} = 0.0013$, $p = 0.92$): returns show no level break either. What the regression does register is the market itself. The S&P 500 return enters with a coefficient of 1.10 ($p < 0.001$)—the familiar market beta near one—and together with the monthly effects it accounts for about a third of the within-firm variation in returns ($R^2 = 0.32$). Volatility adds nothing once the market return is present (VIX, $p = 0.68$). The reading echoes RQ1: firm returns move with the market, as they always have, but the shock itself leaves no separate firm-level step behind. Taken

TABLE II
VARIABLES IN THE RQ1 (STANCE) AND RQ2 (RETURNS) REGRESSIONS.

Variable	Used in	Meaning
bias_{it}	RQ1 (dep.)	Daily firm stance: the signed mean $p_a^+ - p_a^-$ of the headlines about firm i on day t .
return_{it}	RQ2 (dep.)	Daily log return of firm i , $\ln P_{it} - \ln P_{i,t-1}$.
Post_t	RQ1, RQ2	Shock indicator, $1[t \geq 2020-01-01]$; its coefficient β is the parameter of interest.
vol_{it}	RQ1	Coverage volume: the number of retained headlines about firm i on day t .
sp500_ret_t	RQ2	Daily S&P 500 index return, controlling for broad market movement.
vix_t	RQ2	Daily VIX level, controlling for market volatility.
α_i	RQ1, RQ2	Firm effects, absorbing permanent differences across firms.
δ_t^{day}	RQ1	Daily effects, absorbing shocks common to all firms on a day.
δ_t^{month}	RQ2	Monthly effects, absorbing shocks common to all firms in a month.

Note: “(dep.)” marks the dependent variable; the remaining rows are regressors or fixed effects.

TABLE III
REGRESSION RESULTS FOR RQ1 (STANCE, EQ. 2) AND RQ2 (RETURNS, EQ. 3). BOTH FAIL TO REJECT THE NULL OF NO POST-2020 LEVEL SHIFT ONCE COMMON DAILY (RQ1) AND MONTHLY (RQ2) SHOCKS ARE ABSORBED.

(a) RQ1 — Daily Stance (firm + day FE, HC1 SE)			
Variable	Coef.	SE	p
Post ($\hat{\beta}$)	+0.01096	0.02964	0.712
Article volume	−0.00381	0.00049	< 0.001
$n = 27,627$; 26 firms; 3,982 days; within $R^2 = 0.0017$ day-clustered $p_{\text{Post}} = 0.717$			
(b) RQ2 — Daily Returns (firm + month FE, HC1 SE)			
Variable	Coef.	SE	p
Post ($\hat{\beta}$)	−0.00013	0.00132	0.921
S&P 500 return	+1.09876	0.01869	< 0.001
VIX	−0.00003	0.00006	0.684
$n = 20,537$; 26 firms; within $R^2 = 0.316$			

together the two nulls are informative rather than empty—after common shocks are removed, neither the tone of coverage nor firm returns jumped in 2020. A level test is not the only way the shock could have mattered, though: it could have changed how tone and returns move *with one another* over time rather than where each of them sits, and a static regression cannot see that.

C. RQ3:

VI. DISCUSSION

A. RQ1:

B. RQ2:

C. RQ3:

VII. CONCLUSION